

COM2058 Project Phase 1: Data Requirements

Record Store Database System

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1 System Description

This project creates a database for a Record Store. It manages the daily work of the store. The system records albums, artists, store items (vinyl, CDs, cassettes), customers, and orders. It uses a relational database design to keep data safe and prevent repeating information.

2 Entities and Attributes

- **ARTIST:** This is the singer or the music band. It has a unique *ArtistID* (Primary Key), *Name*, *Bio*, and *FormationYear*.
- **ALBUM:** This is the music itself, not the physical item. It has a unique *AlbumID* (Primary Key), *Title*, *ReleaseDate*, and *Genre*.
- **TRACK:** These are the songs on the album. It has a *TrackID* (Primary Key), *Title*, *Duration*, and *TrackNumber*.
- **ALBUM_VARIANT (Physical Inventory):** These are the real items in the store, like CDs or vinyl records. It has a *VariantID* (Primary Key), *Format*, *Price*, and *StockQuantity*.
- **CUSTOMER:** This is the person who buys the music. It has a *CustomerID* (Primary Key), *FirstName*, *LastName*, *Email*, and *Address*.
- **CUSTOMER_ORDER:** This is the shopping receipt or bill. It has an *OrderID* (Primary Key), *OrderDate*, *TotalAmount*, and *Status*.

3 Relationship Requirements

- **Artist and Album (M:N):** One artist can make many albums. Also, one album can have many artists (like a duet). This is a many-to-many relationship.

- **Album and Track (1:N):** One album has many tracks. But one track belongs to only one album.
- **Album and Physical Inventory (1:N):** One album can have many formats (like a CD and a vinyl). But one physical item belongs to only one album.
- **Customer and Order (1:N):** One customer can make many orders. But one order belongs to only one customer.
- **Order and Physical Product (M:N):** One order can have many physical items. Also, a physical item can be in many different orders. This is the *INCLUDES* relationship. It records the *Quantity* and *UnitPrice* at the time of the sale.

4 Business Rules and Constraints

- The *StockQuantity* (number of items in the store) cannot be less than zero.
- Customer *Email* addresses must be unique. Two customers cannot use the same email address.
- The system must save the *UnitPrice* in the order details during the sale. This stops old order prices from changing if the store changes the price of an item later.